3D Random Walk Simulation

示範報告

林致翰　Chih-Han Lin

National Taiwan Univercity,Dept. Of Electrical Engineering

Abstract

Ⅰ. 模擬程式計算流程

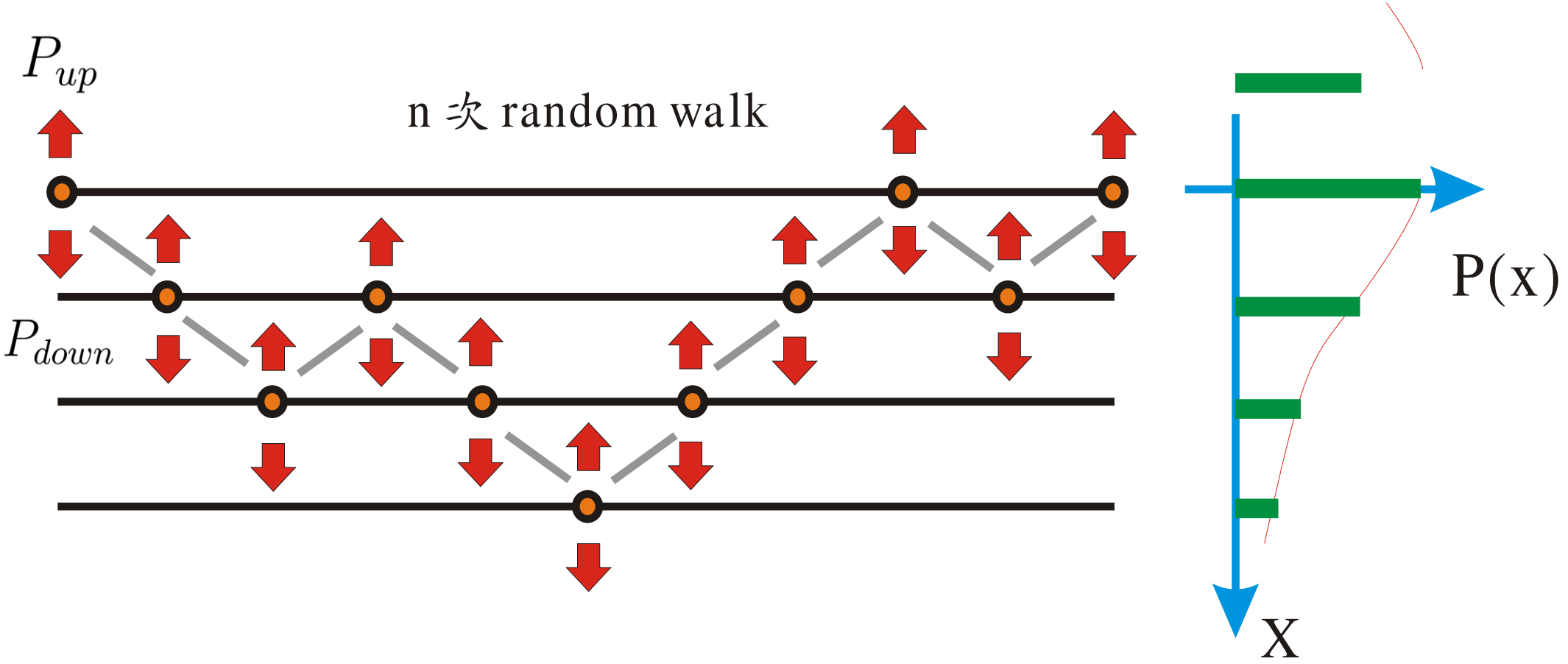
Ⅱ. 自製數據繪圖函數

Ⅲ. 3D Random Walk 模擬

Ⅳ. 程式效能分析

Ⅴ. 結論

Random Walk



假設總共走了 n 步，最後停留在離原點 m 步的位置，則最後出現在 m 處的機率為

$$P_n(m) = \binom{n}{(n+m)/2} \left(\frac{1}{2}\right)^{\frac{n+m}{2}} \left(\frac{1}{2}\right)^{\frac{n-m}{2}}$$

$$P_n(m) = \frac{n!}{(n+m)/2)!(n-m)/2)!} \left(\frac{1}{2}\right)^n$$

$n \to \infty$　則　$P_n(m)$ 會趨近常態分佈

Random Walk 模擬流程

初始化

粒子數 Par_num
模擬時間 simutime
移動間距 step
粒子陣列 Pos(3,par_num)

Pos(3,par_num)

計算粒子移動

呼叫 randomwalk
進行計算，計算結果存至
posf(sam_num,3,par_num)

主要函式:

Randomwalk
drawcum

繪製分佈圖

呼叫 drawcum 整理數據
posf 累次紀錄成 drawdata
以繪製粒子與原點距離之分佈

將 posf 轉換為 normf
(將座標轉換為與原點之距離)

計算函式介紹

```
[Xf,yf,zf]=randomwalk(xi,yi,zi,seed)
```

粒子移動後之座標

粒子原始座標

移動判定所需之亂數

```
for i=1:size(xi,2),
  if seed(1,i)>0.5,
  xf(1,i)=xi(1,i)+step;
else
  xf(1,i)=xi(1,i)-step;
end
```

yf,zf計算方式類推，
seed(1,i) 需改成
seed(2,i)及seed(3,i)

此處 random walk 往兩邊機率相等，可修改判別式以模擬機率不對稱之狀態

Seed 為 3 * par_num 之陣列
xi yi zi 則為 1 * par_num 之陣列

統計繪圖函式介紹

```
drawcum(data,devi_num,maximum,colorf)
```



粒子數據　　計次間距　　數據上界　　繪圖參數

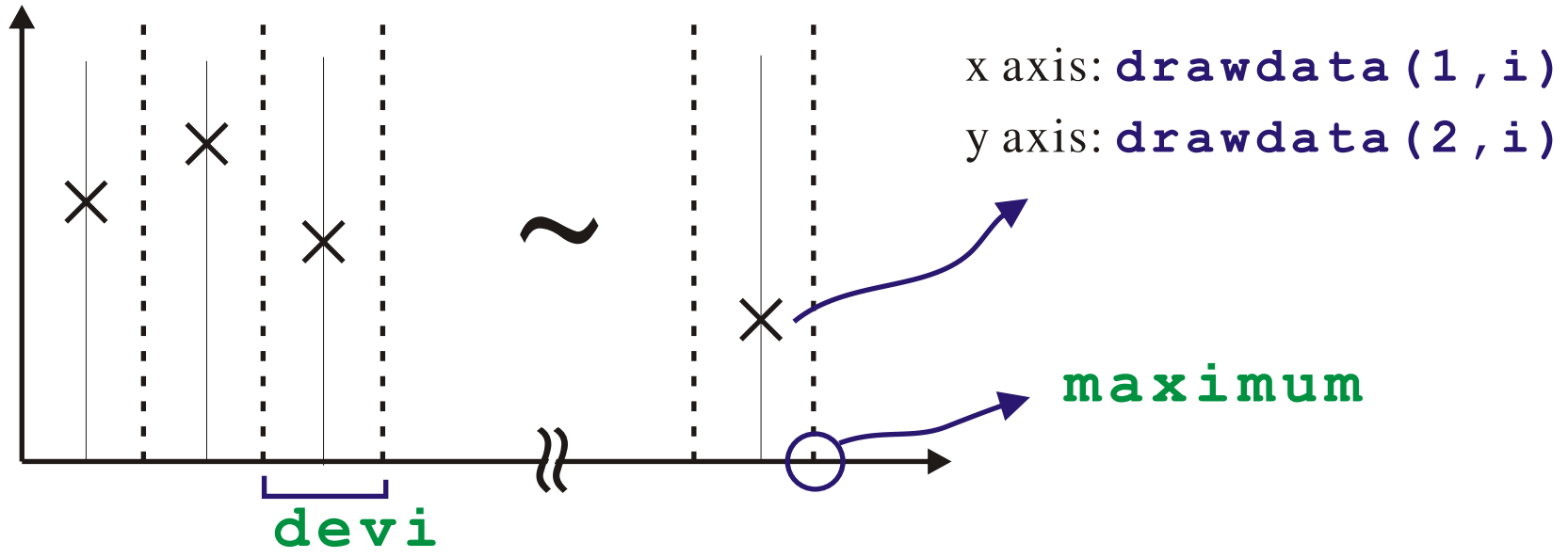
```
drawdata=zeros(2,devi_num+1);
devi=maximum/devi_num;
for i=1:par_num,
  for j=1:devi_num
    if data(1,i)<j*devi,
    drawdata(2,j)=drawdata(2,j)+1;
    end
  end
end
```

由 maximum 與 devi_num 決定統計間距，
drawdata 儲存累積分佈函數（CDF）

```
drawdata(1,1)=0.5*devi;
for k=1:devi_num,
  drawdata(1,k+1)=drawdata(1,k)+devi;
  drawdata(2,devi_num-k+2)=...
  drawdata(2,devi_num-k+2)-...
  drawdata(2,devi_num-k+1);
end
```

將 CDF 改成 PDF

```
plot(drawdata(1,1:devi_num),...
drawdata(2,1:devi_num),colorf);
```



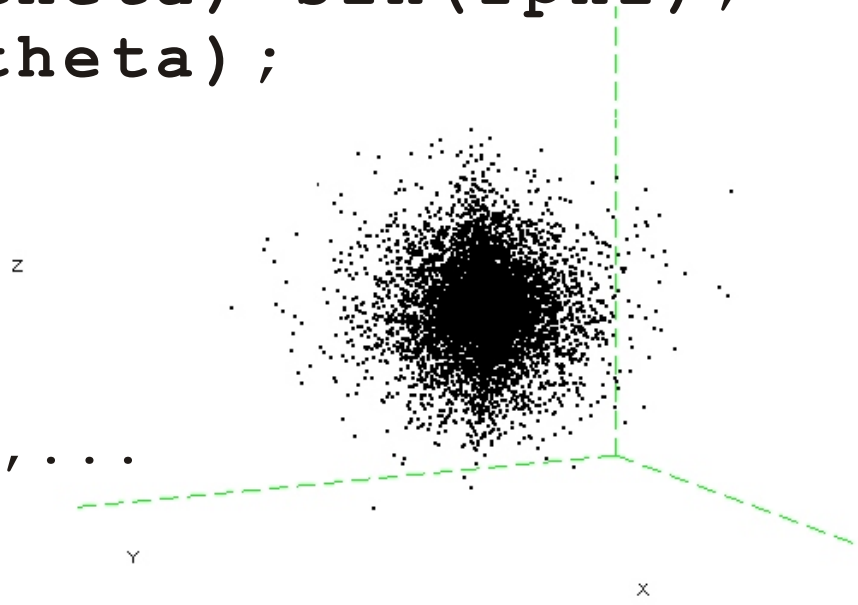
徑向高斯分佈

rand 指令加入 'n' 參數可產生
標準差為 1，平均值為 0 之高斯亂數

```
for i=1:par_num,
  rran=rand(1,1,'n');
  rtheta=rand()*%pi-0.5*%pi;
  rphi=rand()*2*%pi;
  pos(1,i)=rran*cos(rtheta)*cos(rphi);
  pos(2,i)=rran*cos(rtheta)*sin(rphi);
  pos(3,i)=rran*sin(rtheta);
end
```

θ 與 ϕ 方向皆為均勻分佈之亂數

```
param3d1(pos(1,:)',pos(2,:)',...
list(pos(3,:)',0))
```



繪圖函式測試

```
r0=pos.*pos;
for j=1:par_num,
normfi(1,j)=sqrt(sum(r0(:,j)));
end

drawcum(normfi(1,:),100,...
maxi(normfi(1,:)),'mx-');
```

將座標轉換為徑向距離

嘗試代入不同的累計區間數 **devi_num** ，觀察繪圖曲線的平滑程度。
取 **devi_num=20,50,100,200,500** 進行繪製並加以比較

定義 SA (sample ratio) = **devi_num / par_num**

SA = 0.2% , 0.5%, 1%, 2%, 5%

SA 越大則 **drawcum** 執行的時間越長

計時程式

```
run_num=10;
time=zeros(1,run_num);
for i=1:run_num,
tic;
```

置入需計時之程式碼區塊

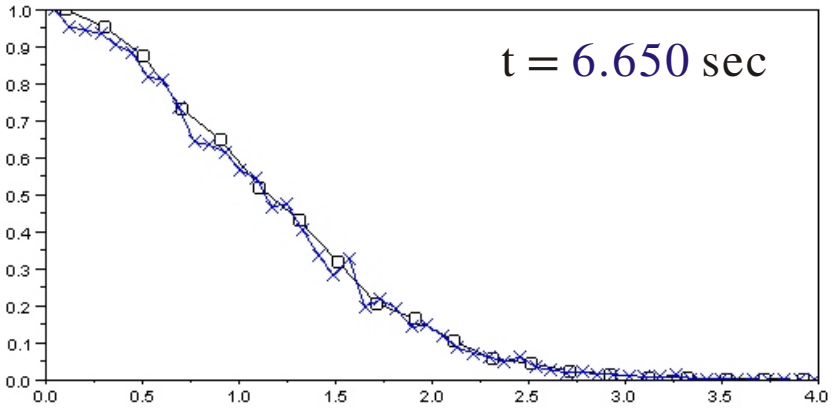
```
toc;
time(i)=ans;
close();
end
avg=sum(time)/size(time,2)
```

於每輪測試完畢自動關閉測試中出現的圖

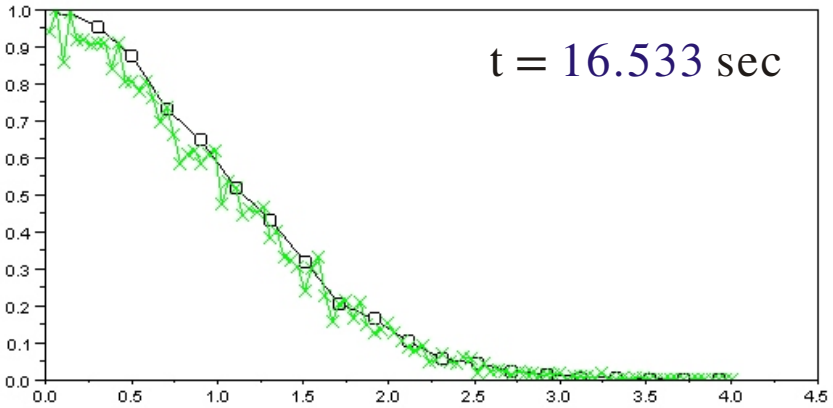
圖中黑色圓點為參考基準 SA=0.2% (t=2.761 sec)

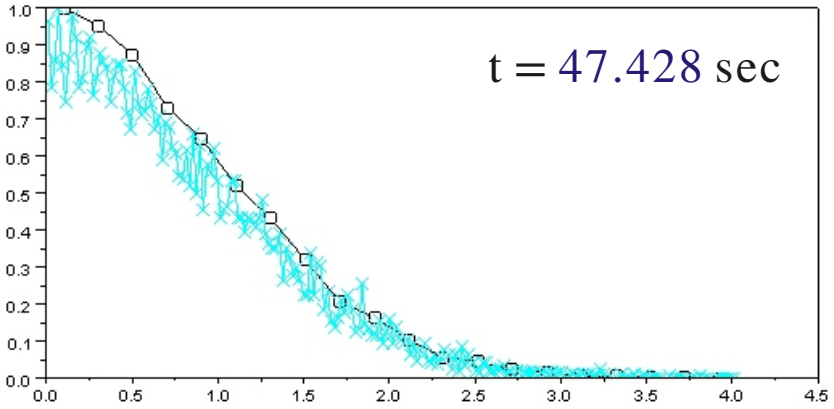
使用 Pentium M 740, 1.73 GHz (~ P4, 3.4 GHz) + 2G DDR2 RAM

SA = 0.5%

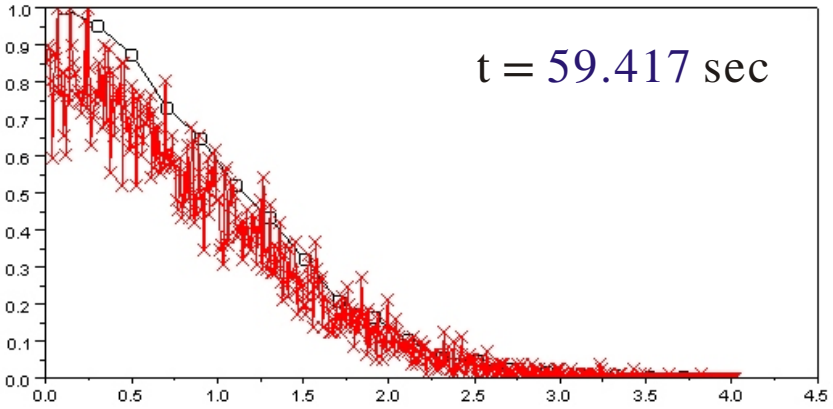


SA = 1 %





SA = 2%



SA = 5%

曲線套適

```
function y=gaussf(x)
  sig2=sqrt(1/(sig^2+10^-6));
  effsig=sqrt(sig2);
  y=effsig/(sqrt(2*%pi))*...
  exp(-(x-ex)^2*sig2/2);
endfunction
```

gaussf(x)
為套適函數，其中
sig 與 ex
為待求係數

$$y(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-(x-\mu)^2}{\sigma}\right]$$

```
function e=G(p,z)
  ex=p(1);
  sig=p(2);
  y=z(1);
  x=z(2);
  e=y-gaussf(x);
endfunction
```

待求係數

原始數據

$$\frac{1}{\sigma} \simeq \sqrt{\frac{1}{\sigma^2 + 10^{-6}}}$$

避免計算產生
divided by zero error

配合 fit_dat 使用之誤差計算函數

```
[pt,et]=fit_dat(G,[0;1],fittingdata);
```

```
p1=zeros(3,100);

wjob=waitbar('fitting...');
for i=1:100,
waitbar(i/100,wjob);
[p1(1:2,i),p1(3,i)]=fit_dat(G,[0;1],...
drawdata);
end
close(wjob);

plot(11:110,p1(1,:),'rx-')
plot(11:110,p1(2,:),'bo-')
```

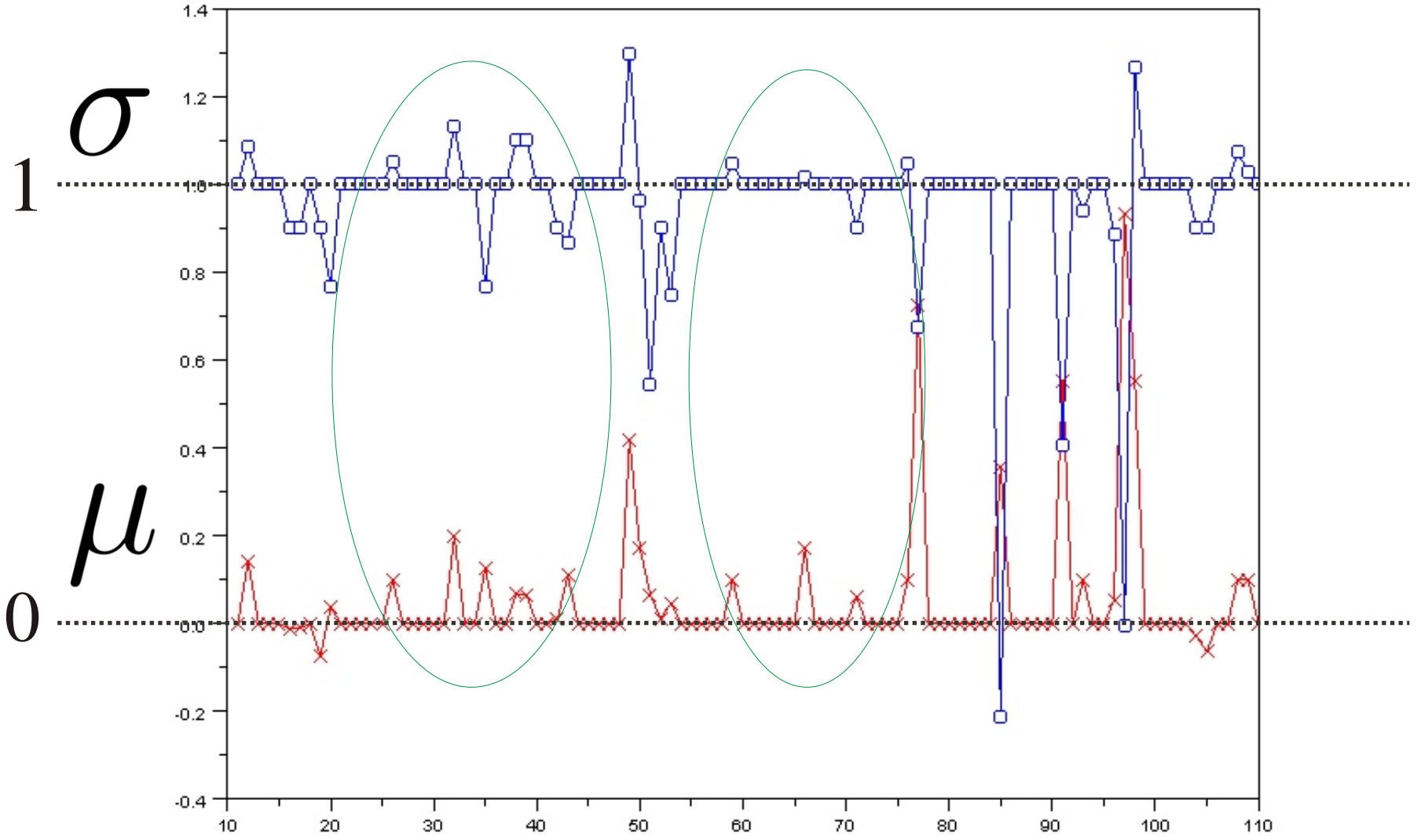
計算 **devi_num** = 11:110 ，透過 **drawcum** 產生的分佈函數**drawdata**是否正確還原成起始的高斯分佈（期望值為 0 ，標準差為 1）

SA = 0.25%~0.45%

25~45

SA = 0.6%~0.75%

60~75



3D Random Walk

```
clear;
par_num=10000;
frame=100;
simutime=1;
devi_num=65;
pos=zeros(3,par_num);
posf=zeros(3,par_num);
step=0.2;


 xtemp=pos(1,:);
 ytemp=pos(2,:);
 ztemp=pos(3,:);
```

```
waitjob1=waitbar('calculate');
for k=1:frame,
  waitbar(k/frame,waitjob1);
  posf(1,:)=xtemp;
  posf(2,:)=ytemp;
  posf(3,:)=ytemp;
  r2=posf.*posf;
  normf=sqrt(r2(1,:)+r2(2,:)+r2(3,:));
  drawcum(normf,devi_num,maxi(normf),'bx-');
  isoview(0,8,0,5);
  filename=string(k)+'.png';
  xs2png(0,filename);
  clf();
    for j=1:simutime,
      [xtemp,ytemp,ztemp]randomwalk...
      (xtemp,ytemp,ztemp,rand(3,par_num));
    end
end
close(waitjob1);
```

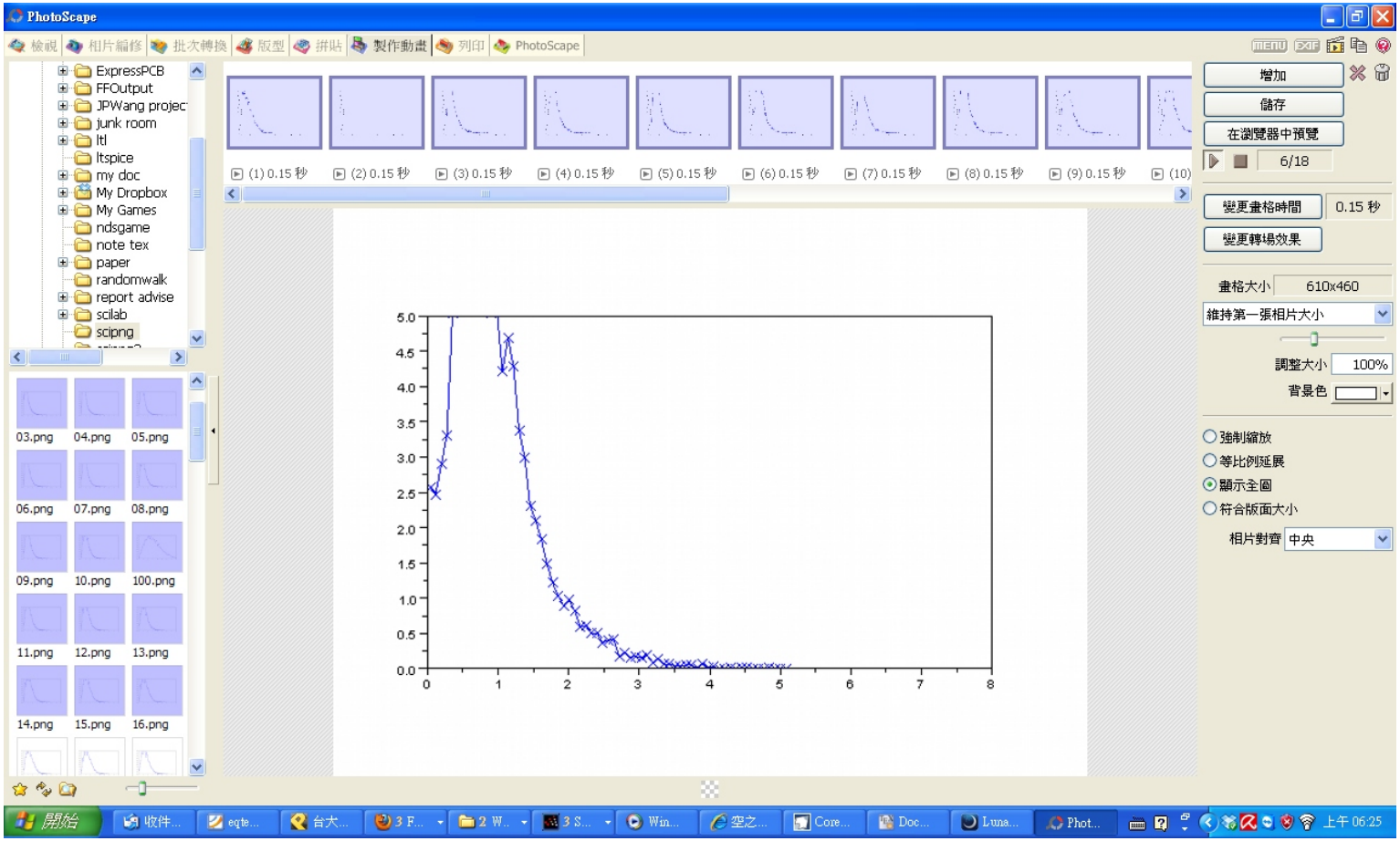
加入waitbar 控管模擬計算進度

外層迴圈將資料輸出成 png 檔（每過 simutime 取一個 frame ）

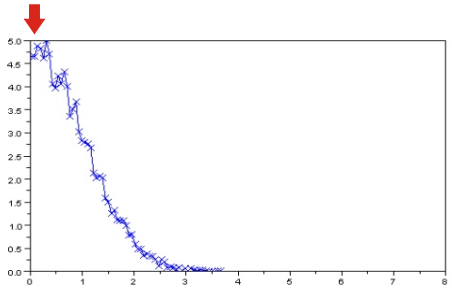
內層迴圈呼叫 randomwalk 算出下個時刻粒子的位置

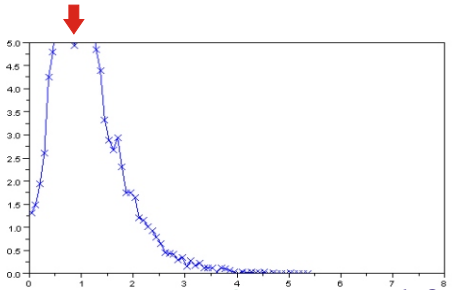
動畫合成

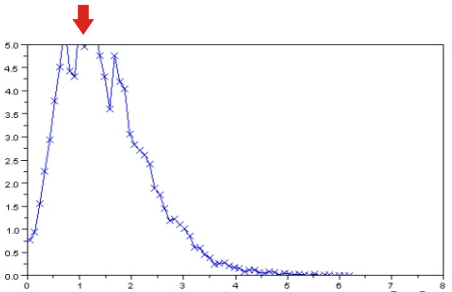
使用 free software ： photoScape 合成影格

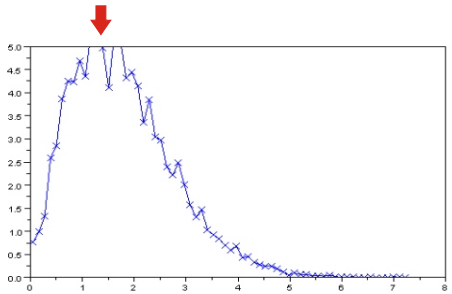


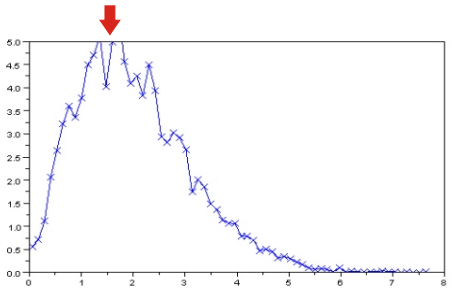
取 simutime=1,frame=100,step=0.2

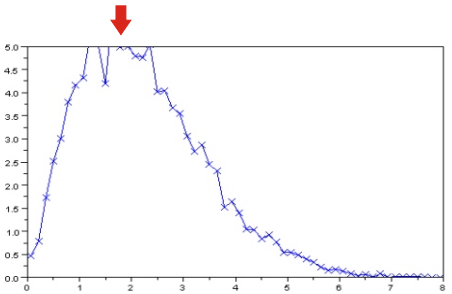

frame=1

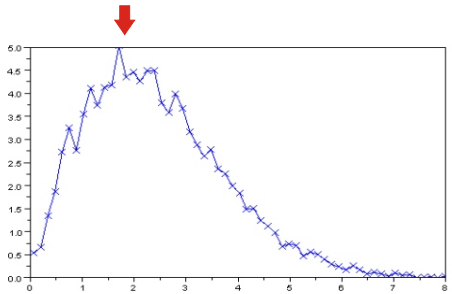

frame=10

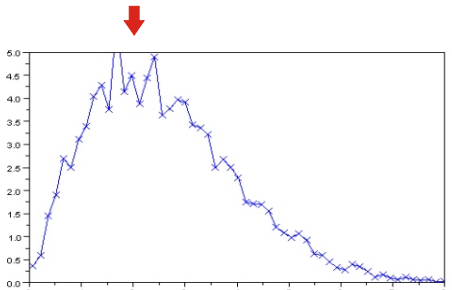

frame=20

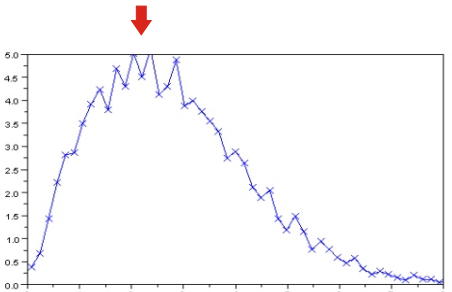

frame=30


frame=40


frame=50

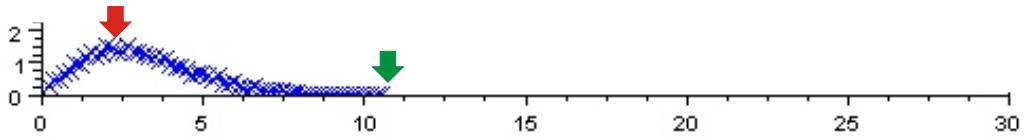

frame=60


frame=70

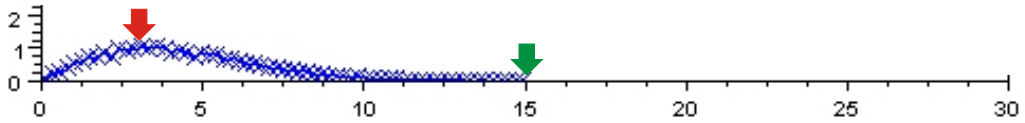

frame=80

取 simutime=1,frame=2000,step=0.2

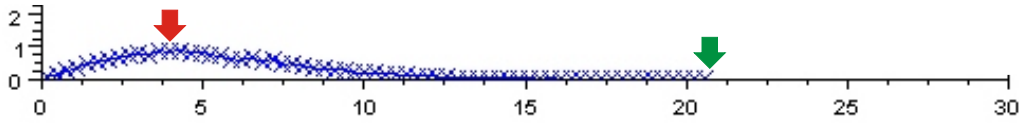
frame=100



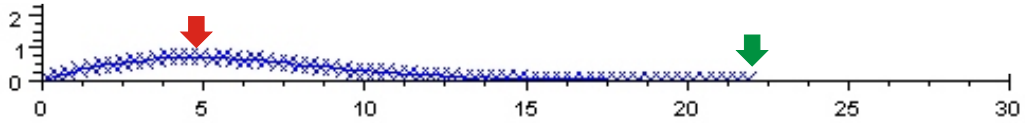
frame=200



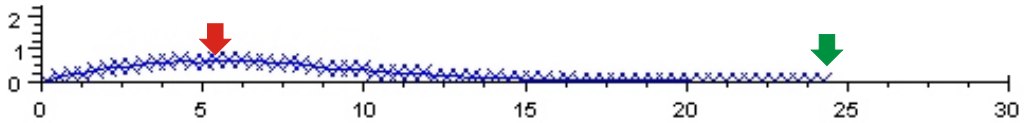
frame=300



frame=400



frame=500



Rayleigh 分佈

The Problem of the Random Walk.

CAN any of your readers refer me to a work wherein I should find a solution of the following problem, or failing the knowledge of any existing solution provide me with an original one? I should be extremely grateful for aid in the matter.

A man starts from a point O and walks l yards in a straight line; he then turns through any angle whatever and walks another l yards in a second straight line. He repeats this process n times. I require the probability that after these n stretches he is at a distance between r and $r+\delta r$ from his starting point, O.

The problem is one of considerable interest, but I have only succeeded in obtaining an integrated solution for *two* stretches. I think, however, that a solution ought to be found, if only in the form of a series in powers of $1/n$, when n is large.

KARL PEARSON.

The Gables, East Ilsley, Berks.

Nature **72**, 294 (1905)

The Problem of the Random Walk.

THIS problem, proposed by Prof. Karl Pearson in the current number of NATURE, is the same as that of the composition of n iso-periodic vibrations of unit amplitude and of phases distributed at random, considered in *Phil. Mag.*, x., p. 73, 1880; xlvii., p. 246, 1899; ("Scientific Papers," i., p. 491, iv., p. 370). If n be very great, the probability sought is

$$\frac{2}{n} e^{-r^2/n} r\,dr,$$

Probably methods similar to those employed in the papers referred to would avail for the development of an approximate expression applicable when n is only moderately great.

RAYLEIGH.

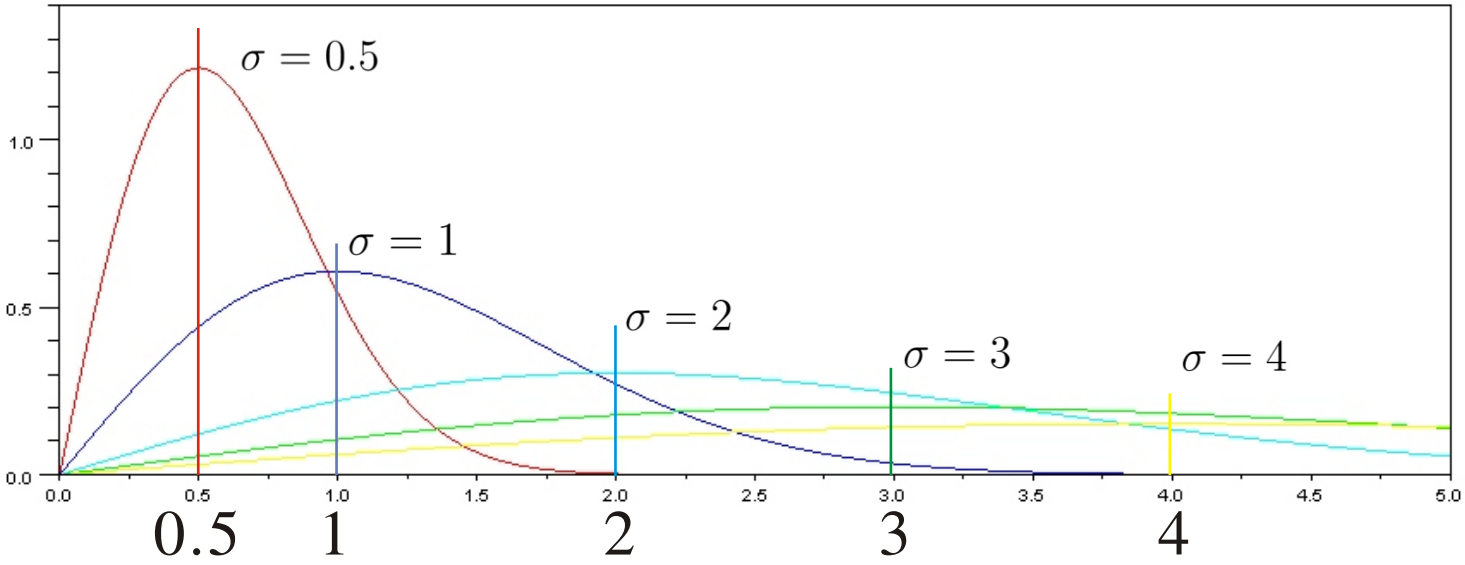
Terling Place, July 29.

Nature 72, 318 (1905)

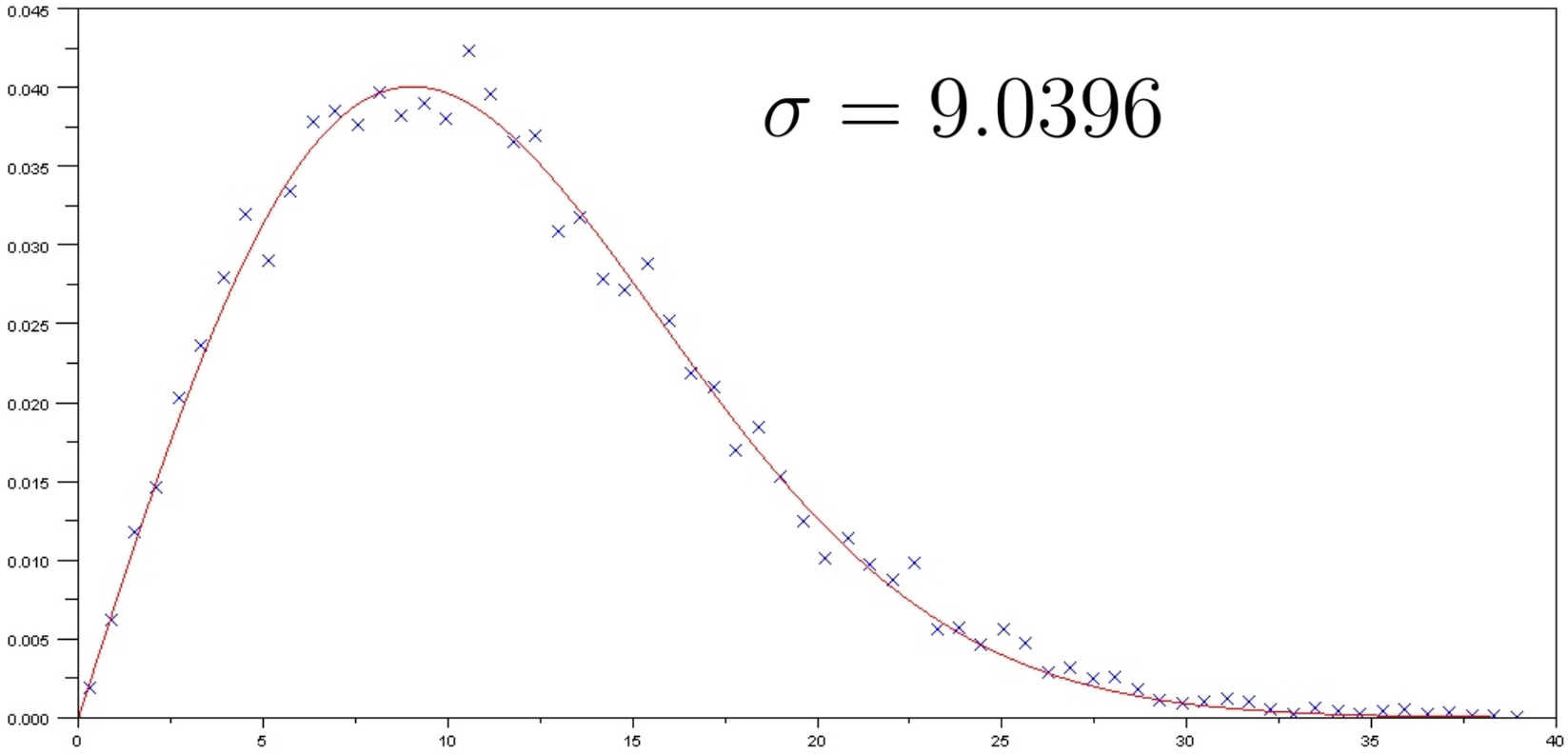
```
function y=rayleigh(x)
  sig2=sqrt(1/(sig^2+10^-6));
  y=x.*maxmut.*(sig2^2).*exp(-x.^2*sig2^2/2);
endfunction
```

$$f_{max} = f(\sigma;\sigma) \simeq \frac{0.606}{\sigma}$$

$$f(x;\sigma) = \frac{x}{\sigma^2}\exp(\frac{-x^2}{2\sigma^2}) \qquad for \quad x \in [0,\infty)$$



取 **frame=1400** 的數據分析並以 rayleigh 分佈套適



分佈尖峰的移動

```
for i=1:600,
  temp=1401:2000;
  filename=string(temp(i))+'.dat';
load(filename);

drawdatao=drawcum2(normf,20,maxi(normf),'b-');
drawdata(1,:)=drawdatao(1,:);
drawdata(2,:)=drawdatao(2,:)/sum(drawdatao(2,:));

p0=[0.5;0.5];et=1;
while et>0.09  do
tic;
p0=p0+[rand();3*rand()];
[p0,et]=fit_dat(G,p0,drawdata);
toc;
printf('frame= %d ,sigma= %f,maxmut=%f...
err= %f,cputime=%f\n',i,p0(1),p0(2),et,ans);
end
```

將函式改寫成不執行畫圖僅輸出排序後之陣列

先隨機挑幾筆資料套適
找出合理的 et 值

即時輸出運算概況

```
sig_array(1,i)=p0(1);
maxmut_array(1,i)=p0(2);
errorbar(1,i)=et;
sig=p0(1);
maxmut=p0(2);
clear normf;
end
```

紀錄各 frame 套適後所得之參數

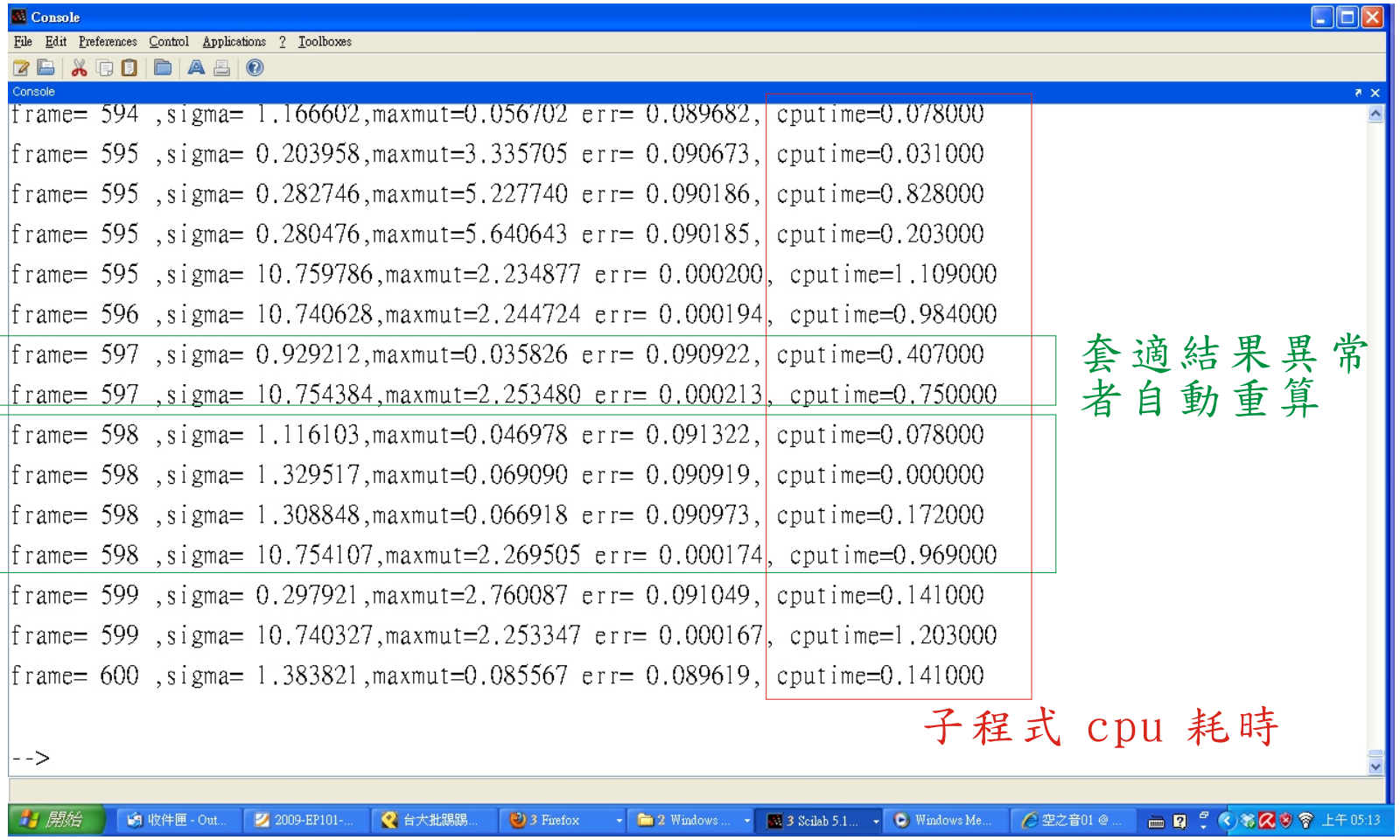
```
function y=linearfit(x)
  y=a*x+b;
endfunction
```

線性套適

```
function e=Gl(p,z)
  a=p(1);
  b=p(2);
  y=z(2);
  x=z(1);
  e=y-linearfit(x);
endfunction
```

對應的 error funciotn

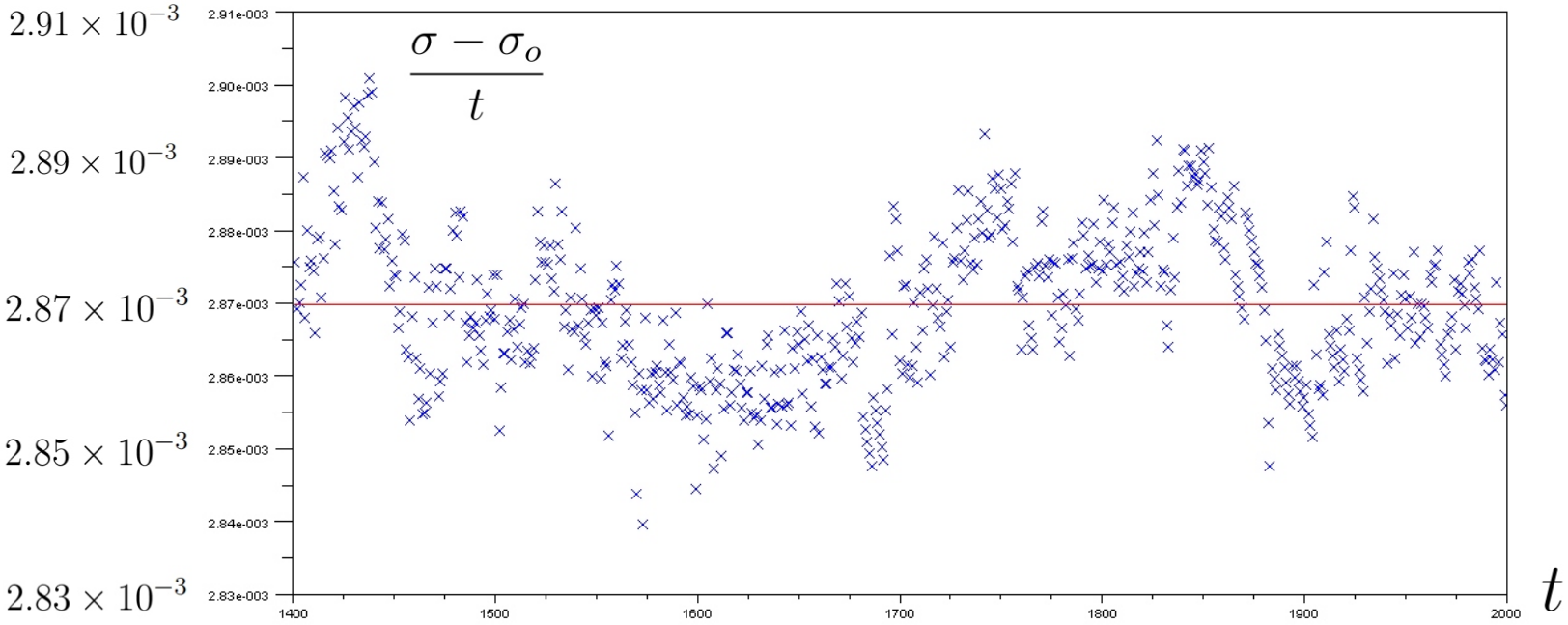
程式運算過程


Console
File Edit Preferences Control Applications ? Toolboxes
Console
frame= 594 ,sigma= 1.166602,maxmut=0.056702 err= 0.089682, cputime=0.078000
frame= 595 ,sigma= 0.203958,maxmut=3.335705 err= 0.090673, cputime=0.031000
frame= 595 ,sigma= 0.282746,maxmut=5.227740 err= 0.090186, cputime=0.828000
frame= 595 ,sigma= 0.280476,maxmut=5.640643 err= 0.090185, cputime=0.203000
frame= 595 ,sigma= 10.759786,maxmut=2.234877 err= 0.000200, cputime=1.109000
frame= 596 ,sigma= 10.740628,maxmut=2.244724 err= 0.000194, cputime=0.984000
frame= 597 ,sigma= 0.929212,maxmut=0.035826 err= 0.090922, cputime=0.407000
frame= 597 ,sigma= 10.754384,maxmut=2.253480 err= 0.000213, cputime=0.750000
frame= 598 ,sigma= 1.116103,maxmut=0.046978 err= 0.091322, cputime=0.078000
frame= 598 ,sigma= 1.329517,maxmut=0.069090 err= 0.090919, cputime=0.000000
frame= 598 ,sigma= 1.308848,maxmut=0.066918 err= 0.090973, cputime=0.172000
frame= 598 ,sigma= 10.754107,maxmut=2.269505 err= 0.000174, cputime=0.969000
frame= 599 ,sigma= 0.297921,maxmut=2.760087 err= 0.091049, cputime=0.141000
frame= 599 ,sigma= 10.740327,maxmut=2.253347 err= 0.000167, cputime=1.203000
frame= 600 ,sigma= 1.383821,maxmut=0.085567 err= 0.089619, cputime=0.141000
-->
套適結果異常者自動重算
子程式 cpu 耗時
開始
收件匣 - Out...
2009-EP101-...
台大批踢踢...
3 Firefox
2 Windows ...
3 Scilab 5.1...
Windows Me...
空之音01 @ ...
上午 05:13

```
pt2=[1;1];
avg2(1,:)=1401:2000;
avg2(2,:)=sig_array;
[pt2,et2]=fit_dat(Gl,pt2,avg2(1:2,:));
```

$$\sigma = kt + \sigma_0$$

$$k = 2.8698 \times 10^{-3}$$
$$\sigma_0 = 5.0283$$



$$\mu(X) = \sigma\sqrt{\frac{\pi}{2}} \simeq 1.253\ \sigma$$

已知 Rayleigh 分佈的期望值與 σ 的關係

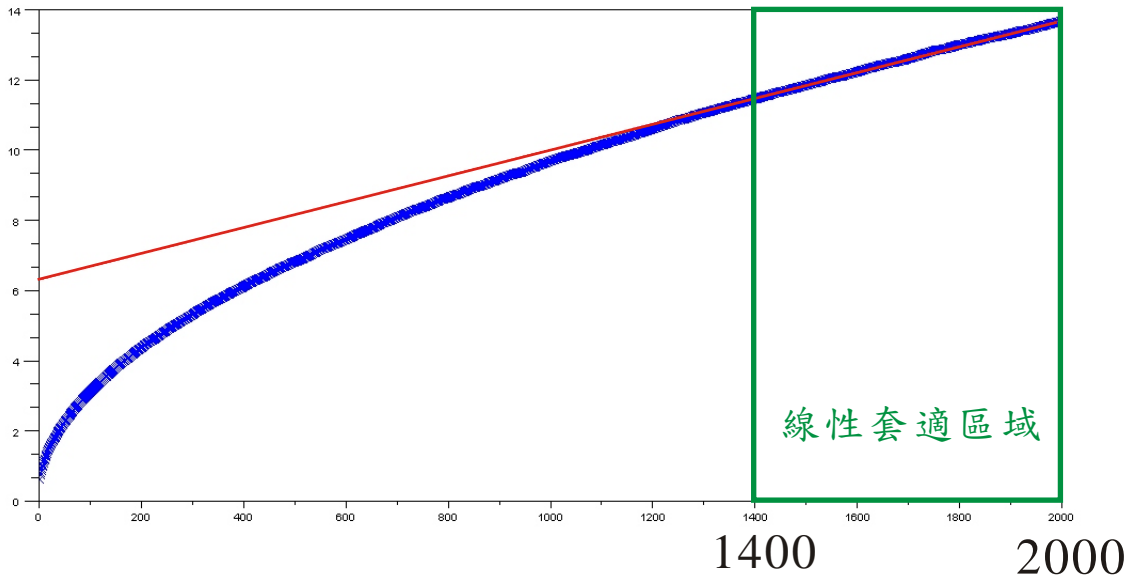
直接分析原始資料的平均值可看出整個分佈如何從從高斯分佈演變成 Rayleigh 分佈

$$\mu = k_\mu t + \mu_0$$

$$k = k_\mu / 1.253$$

$$k_\mu = 3.6981 \times 10^{-3}$$

$$k = 2.9513 \times 10^{-3}$$



擴散模型的應用

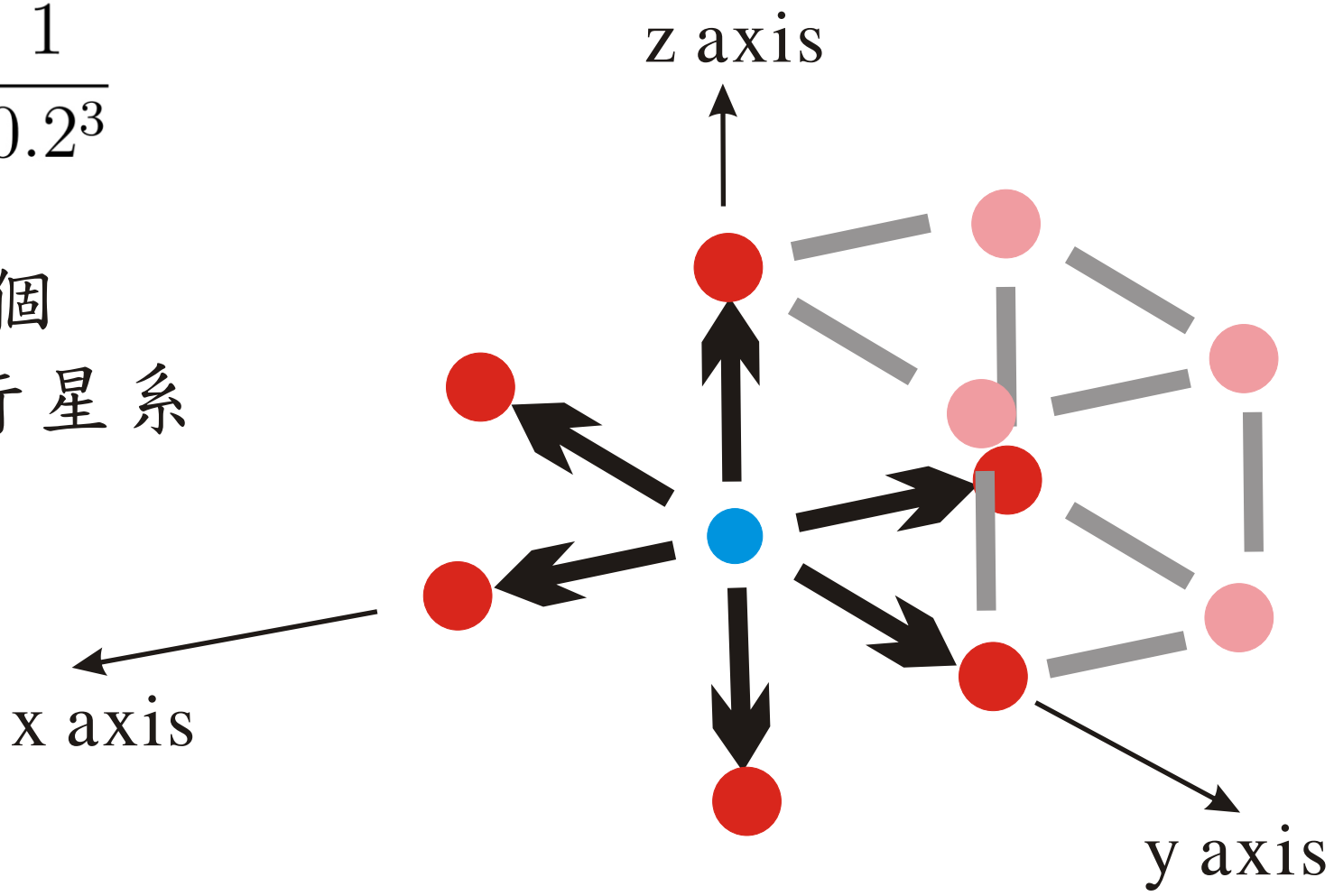
如果我想向 Asimov 致敬，書寫一部銀河帝國興亡史...

Brownian Galaxy Colonist

假設某個種族位於銀河系中心，人口呈現高斯分佈，佔據數光年的星域；該種族的科技足以讓他們往四面八方侵略，若該種族的以N年為遷移週期，遷移步距 0.2 光年採取 Ramdom Walk 的形式擴張...

$$\frac{4\pi}{3}(10^4)^3\frac{1}{0.2^3}$$

~ 5.2 億個
可殖民行星系



假設可殖民之行星系呈立方晶格排列
晶格長度為 0.2 光年

無論起始人口如何分佈，最後總是形成 Rayleigh 分佈；帝國由中央集權變為分裂在以 3×10^{-3} 的速度往銀河系外側推進的殼狀聯邦 . . .

最外圍移至 1 萬光年須耗時(T1):

$$\frac{1\text{ 萬}}{0.2} \times \text{遷移週期（年）}$$

人口最密集處轉移到 1 萬光年須耗時（T2）:

$$\frac{1\text{ 萬}}{3 \times 10^{-3}} \times \text{遷移週期（年）}$$

取遷移週期 ～200 年

（假設載具運量佔人口 0.5％，
載具移動速度為 0.4 倍光速）

T1 ＝ 一千萬 年

T2＝六億七千萬 年

結論與展望

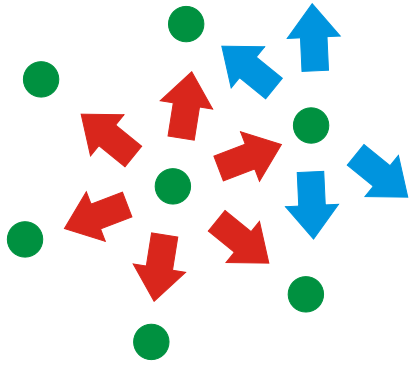
Random Walk 的結果與 Rayleigh distribution 和 chi-square distribution 有緊密的關係

僅僅是利用程式擲了數千萬次的硬幣並加以分析，

我們卻可以將其類比為某些簡單情況下的擴散行為

如何加入自我複製的特性來模擬 Von Neumann Probe？

2 維六方晶格的 Random Walk？



Reference

Wiki Pedia 中 Rayleigh distribution, chi-square distribution 條 目

MIT OPENCOURSEWARE 18.366 Random Walks and Diffusion Lecture notes

Statical Mechanics 2ed , Pathria , , Chap.14

Nature **72**, 318 (1905)

Nature **72**, 294 (1905)

Thanks for Your Attention!